TERRAFORM PRESETUP

- Terraform is an open-source Infrastructure as Code (IaC) tool from HashiCorp, allowing users to define, provision, and manage cloud (AWS, Azure, GCP) and on-premises infrastructure using a declarative language (HCL) in configuration files, enabling automation, versioning, and efficient, consistent management of resources like VMs, networks, and storage across different environments.

KEY CONCEPT:

- **Infrastructure as Code (IaC)**: Describes infrastructure (servers, databases, etc.) in code, treating it like software for version control and automation.
- **HashiCorp Configuration Language (HCL)**: Terraform's human-readable language for writing infrastructure definitions.
- **Providers**: Plugins that allow Terraform to interact with various platforms (AWS, Azure, VMware, etc.) via their APIs.
- **Declarative Approac** : You describe the *desired state* of your infrastructure, and Terraform figures out *how* to achieve it.

HOW ITS WORKS:

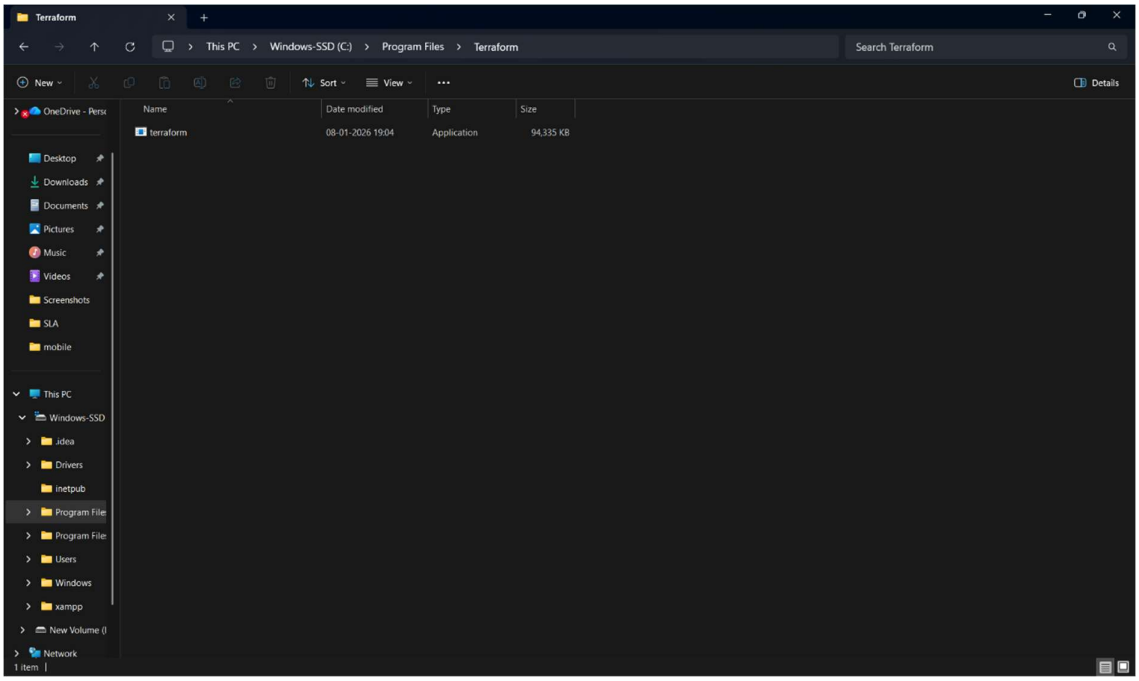
- **Define**: Write configuration files (e.g., .tf) describing your infrastructure.
- **Plan**: Terraform generates an execution plan showing what changes it will make.
- **Apply**: Terraform provisions, updates, or destroys resources to match your desired state.
- **State Management**: Terraform maintains a state file to track resources and detect drift.

BENEFITS

- **Automation**: Automates complex infrastructure provisioning.
- **Consistency**: Ensures environments are identical across development, staging, and production.

- **Multi-Cloud**: Manages resources across multiple providers from a single tool.
- **Versioning**: Infrastructure configurations can be version-controlled in Git.
- **Reusability**: Use modules to create standardized, reusable components

INSTALLATION OF TERRAFORM:
To install an terraform an terraform.io website to click install an latest version of an terraform.

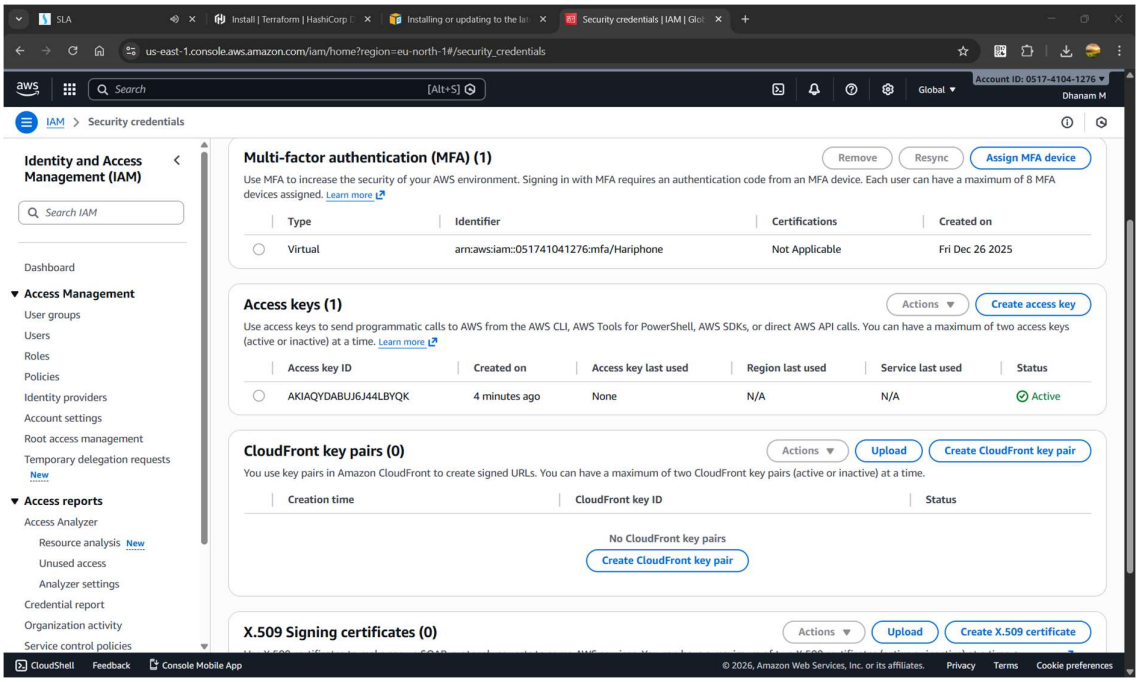


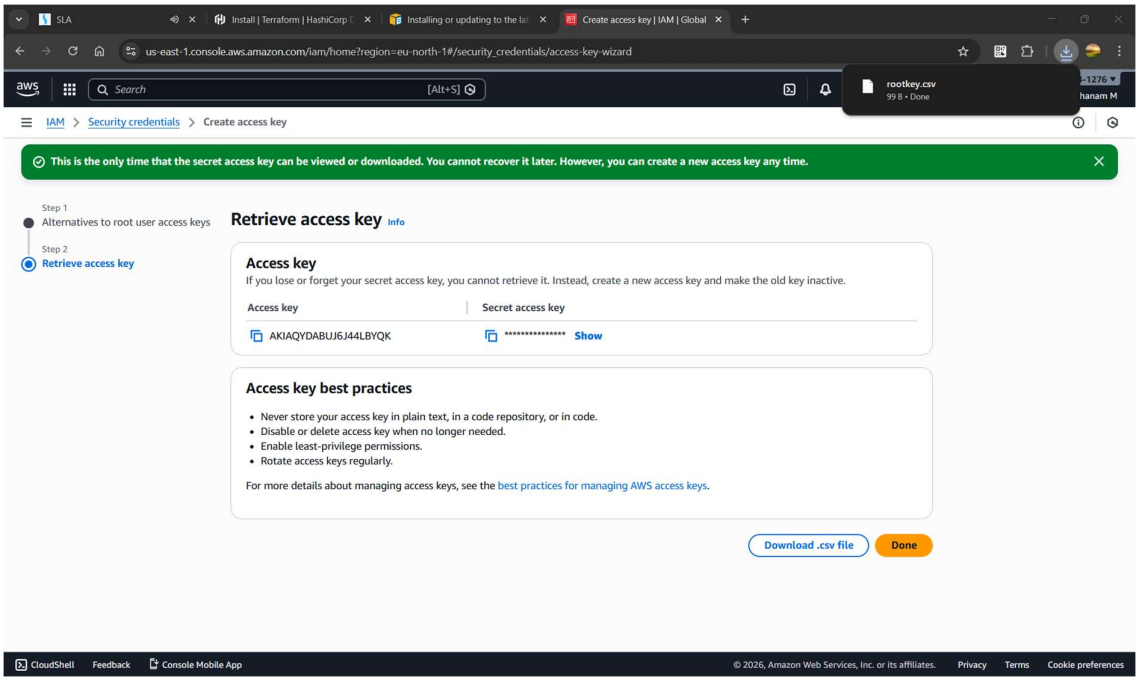
INSTALLATION OF AWS CLI:
To search an aws cli install in google chrome to click an website to install an latest version an aws cli.

```
Command Prompt
Microsoft Windows [Version 10.0.26100.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hariharan>aws --version
aws-cli/2.32.31 Python/3.13.11 Windows/11 exe/AMD64

C:\Users\hariharan>aws configure
```

To login an CLI need an access key and secret acess key of an aws account. Create an access key and secret access key.

```
C:\Users\hariharan>aws configure
AWS Access Key ID [None]:
AWS Secret Access Key [None]:
```




SLA
Install | Terraform | HashiCorp
Installing or updating to the lat
Create access key | IAM | Global
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/security_credentials/access-key-wizard
rootkey.csv
99 B • Done
aws
Search
[Alt+S]
IAM > Security credentials > Create access key
This is the only time that the secret access key can be viewed or downloaded. You cannot recover it later. However, you can create a new access key any time.
Step 1
Alternatives to root user access keys
Step 2
Retrieve access key
Retrieve access key Info
Access key
If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.
Access key
Secret access key
AKIAQYDABUJ6J44LBYQK
********** Show
Access key best practices
Never store your access key in plain text, in a code repository, or in code.
Disable or delete access key when no longer needed.
Enable least-privilege permissions.
Rotate access keys regularly.
For more details about managing access keys, see the best practices for managing AWS access keys.
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Feedback
Console Mobile App
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